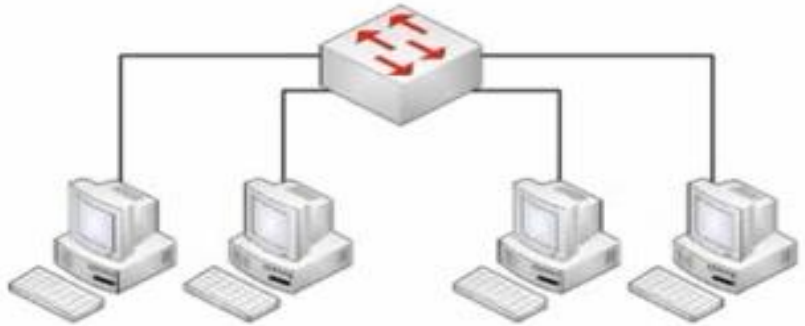
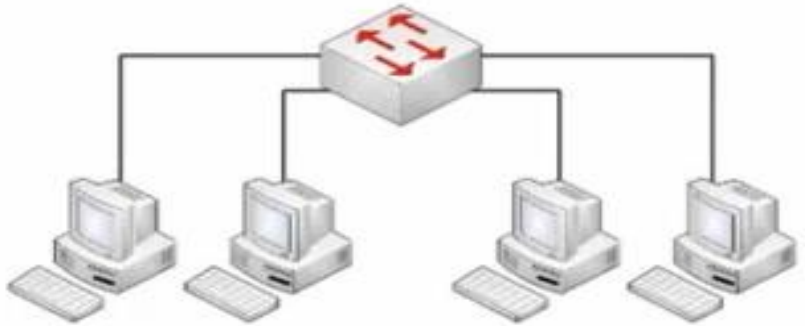


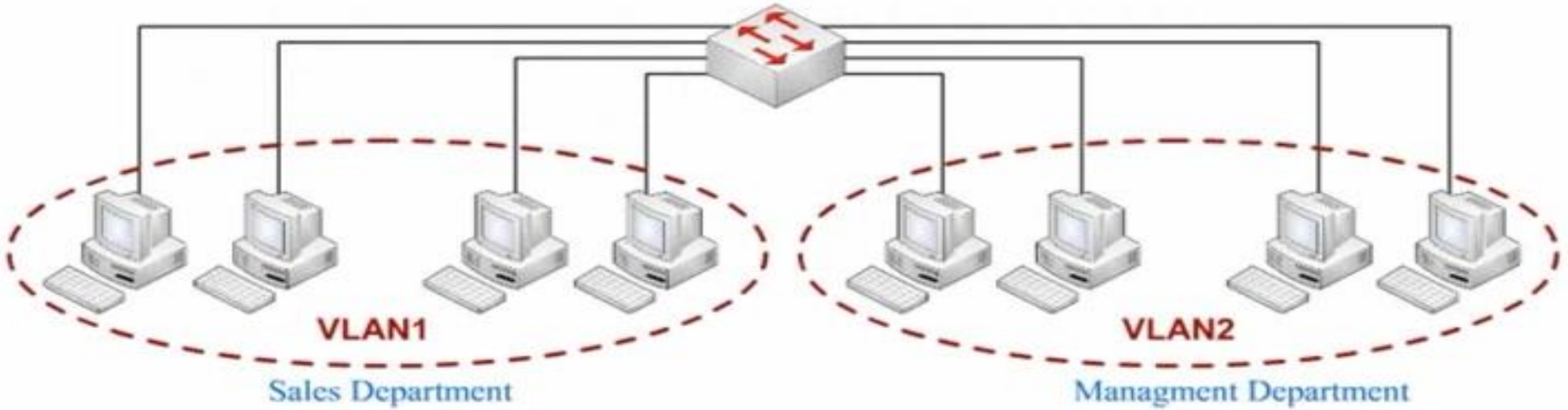
VLAN
Concepts and configuration



Sales Department



Managment Department

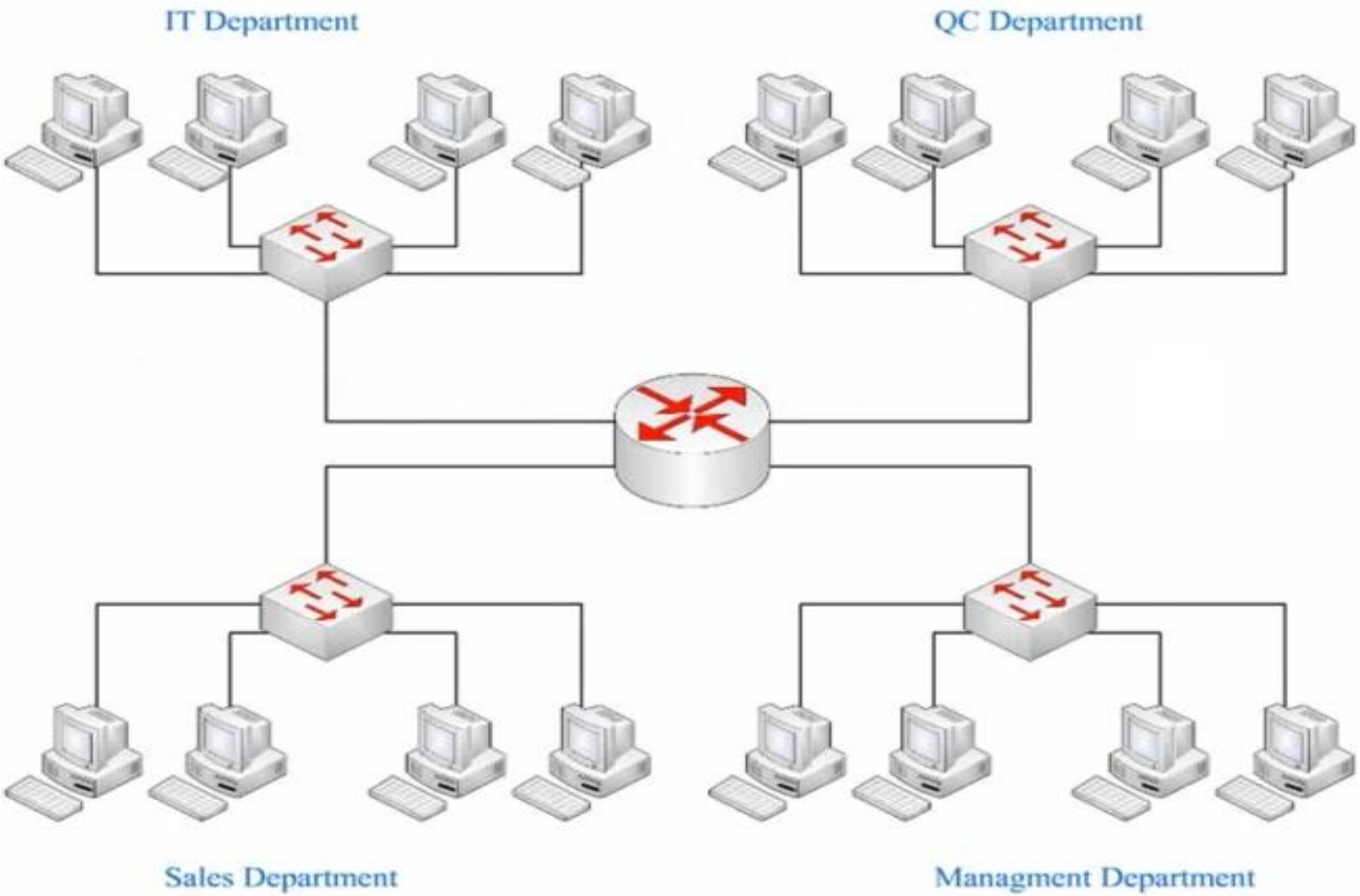


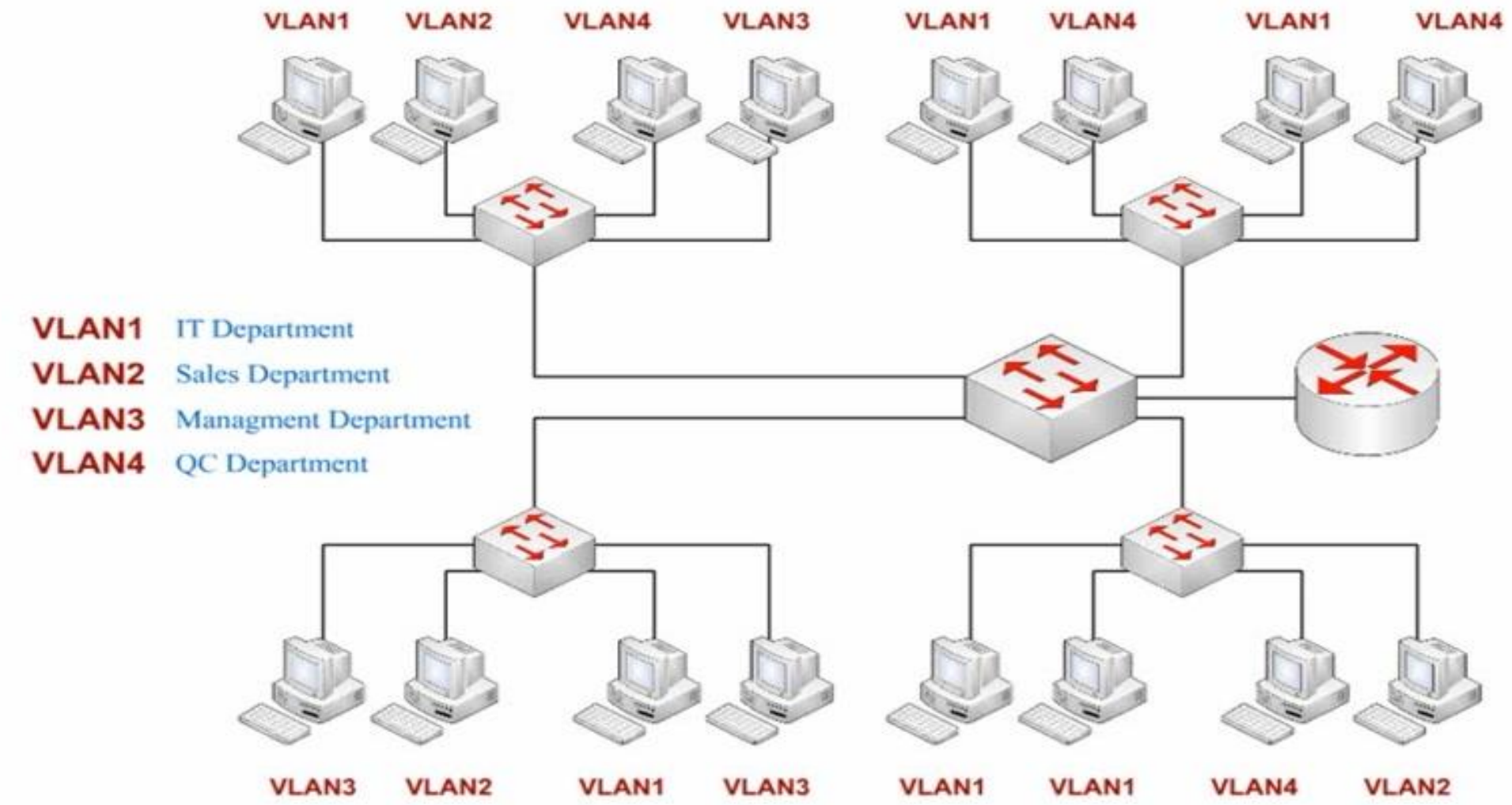
Sales Department

Managment Department

VLAN is a logical way of grouping clients and resources connected to a switch ports.

It's a way of breaking a large broadcast domain up in a layer 2 switching network.


IT Department
QC Department
Sales Department
Managment Department


VLAN1
VLAN2
VLAN4
VLAN3
VLAN1
VLAN4
VLAN1
VLAN4
VLAN1 IT Department
VLAN2 Sales Department
VLAN3 Managment Department
VLAN4 QC Department
VLAN3
VLAN2
VLAN1
VLAN3
VLAN1
VLAN1
VLAN4
VLAN2

VLAN Numbers

- 1-1005 (normal VLAN)
- 1005-4096 (extended VLAN)

VLAN 1 default
VLAN 1002 fddi-default
VLAN 1003 token-ring-default
VLAN 1004 fddinet-default
VLAN 1005 trnet-default

VLAN Configuration

Switch(config)# Vlan *<vlan-id>*
Switch(config-vlan)# name *<name>* Optional

Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan *<vlan-id>*

Switch# show vlan brief
Switch# show mac-address-table vlan *<vlan-id>*

VTP



VTP (VLAN Trunking Protocol)
Switching

VTP Modes

- VTP Server
- VTP Client
- VTP Transparent

VTP

VTP Modes

VTP Server

- VLAN Management
  (Add , Rename & delete)

- Makes VTP advertisements and forward them on all active Trunk ports
  (every 5 minutes and whenever a change occurs)

1- Generate
2- Process
3- Forward

VTP

VTP Modes

VTP Server

VTP advertisements

- VTP Domain
- VLAN Number
- VLAN Name
- VTP Version
- VTP Revision number
- VTP Password
- ...

VTP Modes

VTP Server

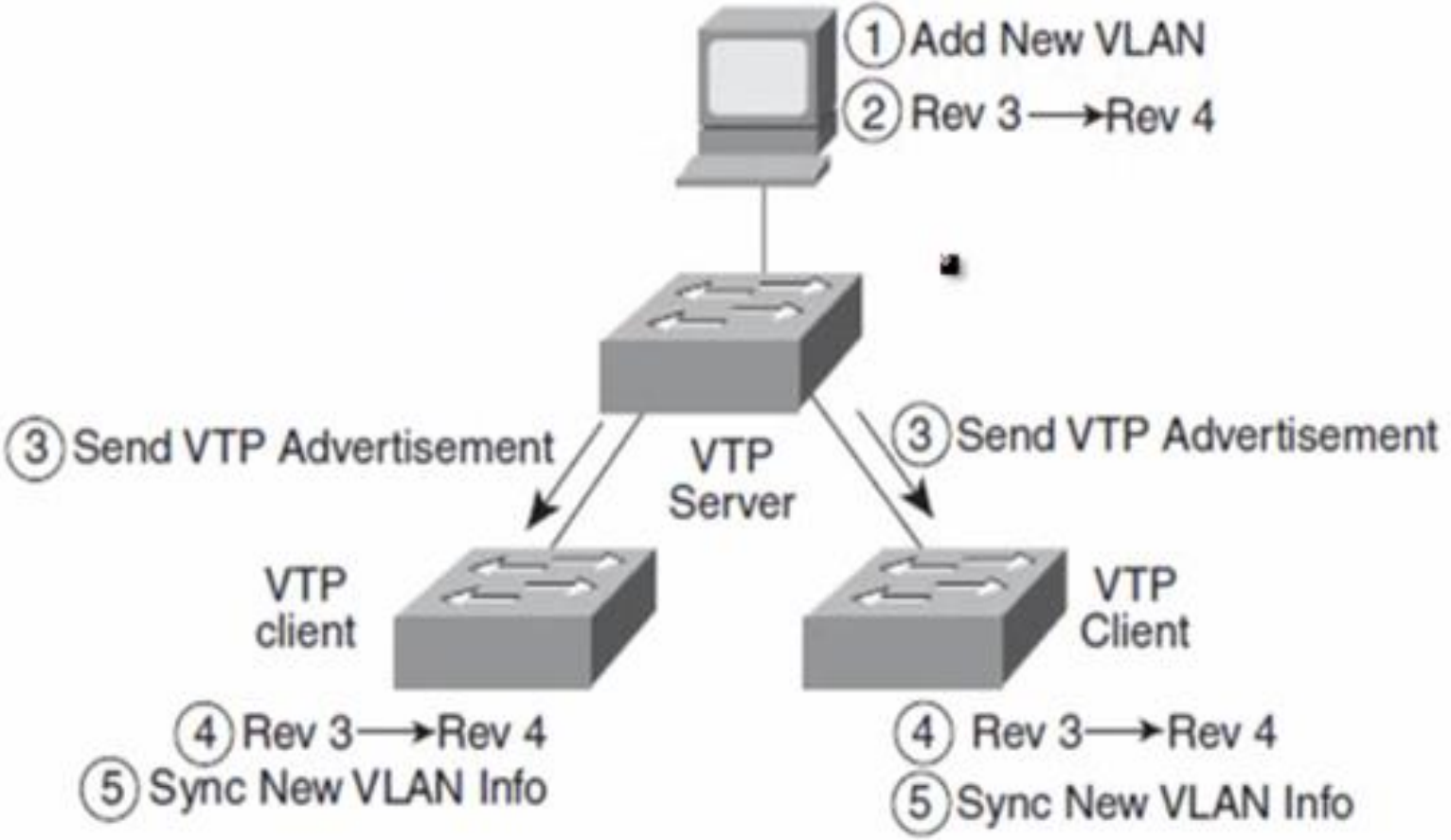
Note

- VTP Servers are VTP Clients too (Redundancy).
- Switch ignores advertisements that have a different *VTP Domain name* or an earlier *configuration revision number.*

VTP Modes

VTP Server

Configuration Revision Number



VTP Modes

VTP Client

- No access to modify VLANs.
- They receive , process and forward (on trunk interfaces) VTP advertisements.

VTP Modes

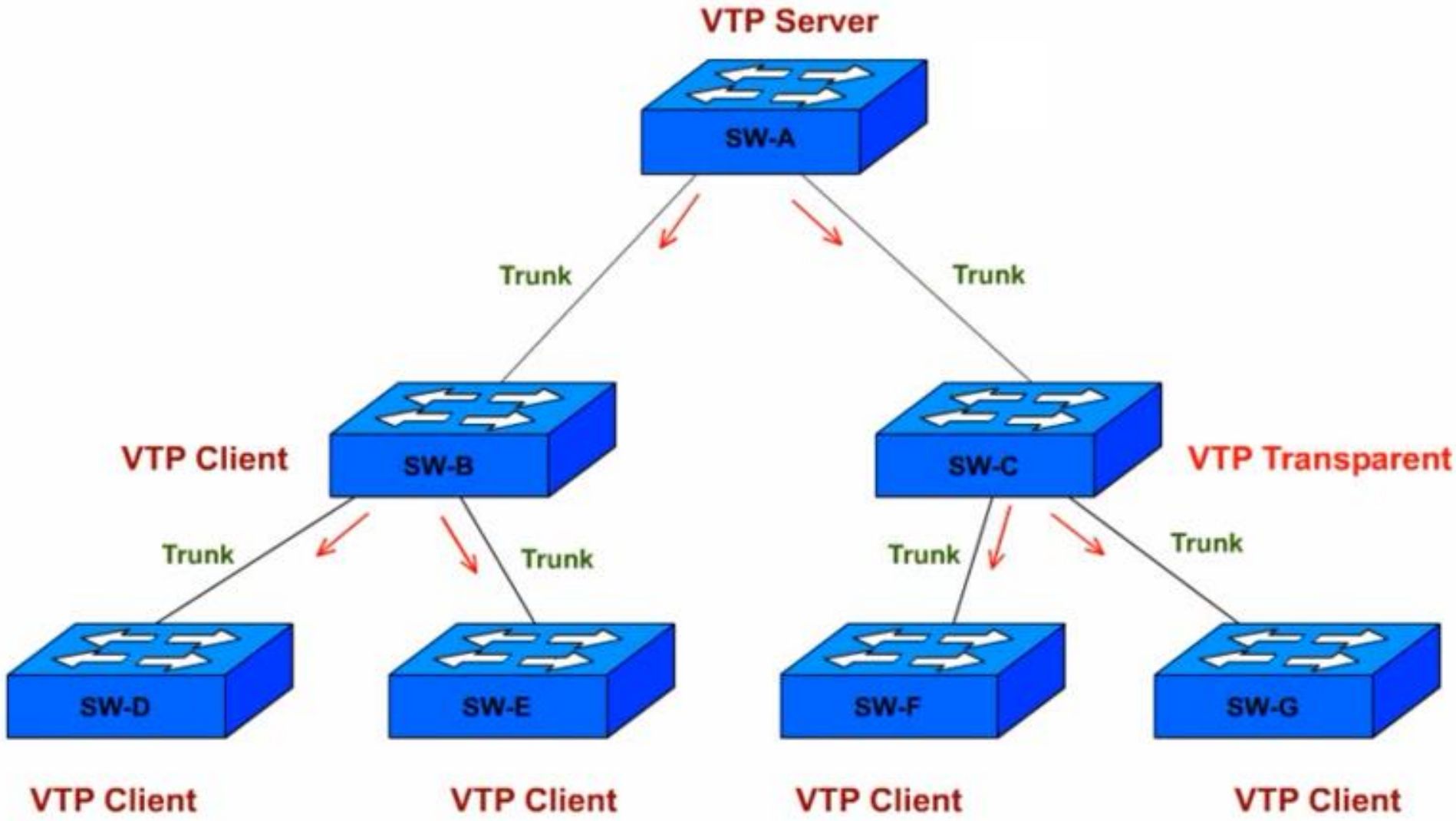
VTP Transparent

- Independent switches that can modify local VLANs.
- They just forward (on trunk interfaces) VTP advertisements with no process.

VTP Modes

Function	Server	Client	Transparent
Only sends VTP messages out ISL or 802.1Q trunks	Yes	Yes	Yes
Supports CLI configuration of VLANs	Yes	No	Yes
Can use normal-range VLANs (1–1005)	Yes	Yes	Yes
Can use extended-range VLANs (1006–4095)	No	No	Yes
Synchronizes (updates) its own config database when receiving VTP messages with a higher revision number	Yes	Yes	No
Creates and sends periodic VTP updates every 5 minutes	Yes	Yes	No
Does not process received VTP updates but does forward received VTP updates out other trunks	No	No	Yes
Places the VLAN ID, VLAN name, and VTP configuration into the running-config file	No	No	Yes
Places the VLAN ID, VLAN name, and VTP configuration into the vlan.dat file in flash	Yes	Yes	Yes

VTP


VTP Server
SW-A
Trunk
Trunk
VTP Client
SW-B
SW-C
VTP Transparent
Trunk
Trunk
Trunk
Trunk
SW-D
SW-E
SW-F
SW-G
VTP Client
VTP Client
VTP Client
VTP Client

VTP consideration

- At least one VTP server must exist.
- Links between switches must be Trunk.
- Trunk links should have a same encapsulation.
- VTP Domain should be the same on all the switches.
- VTP Password (if exists) should be the same on all the switches.
- VTP Version is not important to be the same.

Where VTP Clients and Servers Store VLAN-Related Configuration

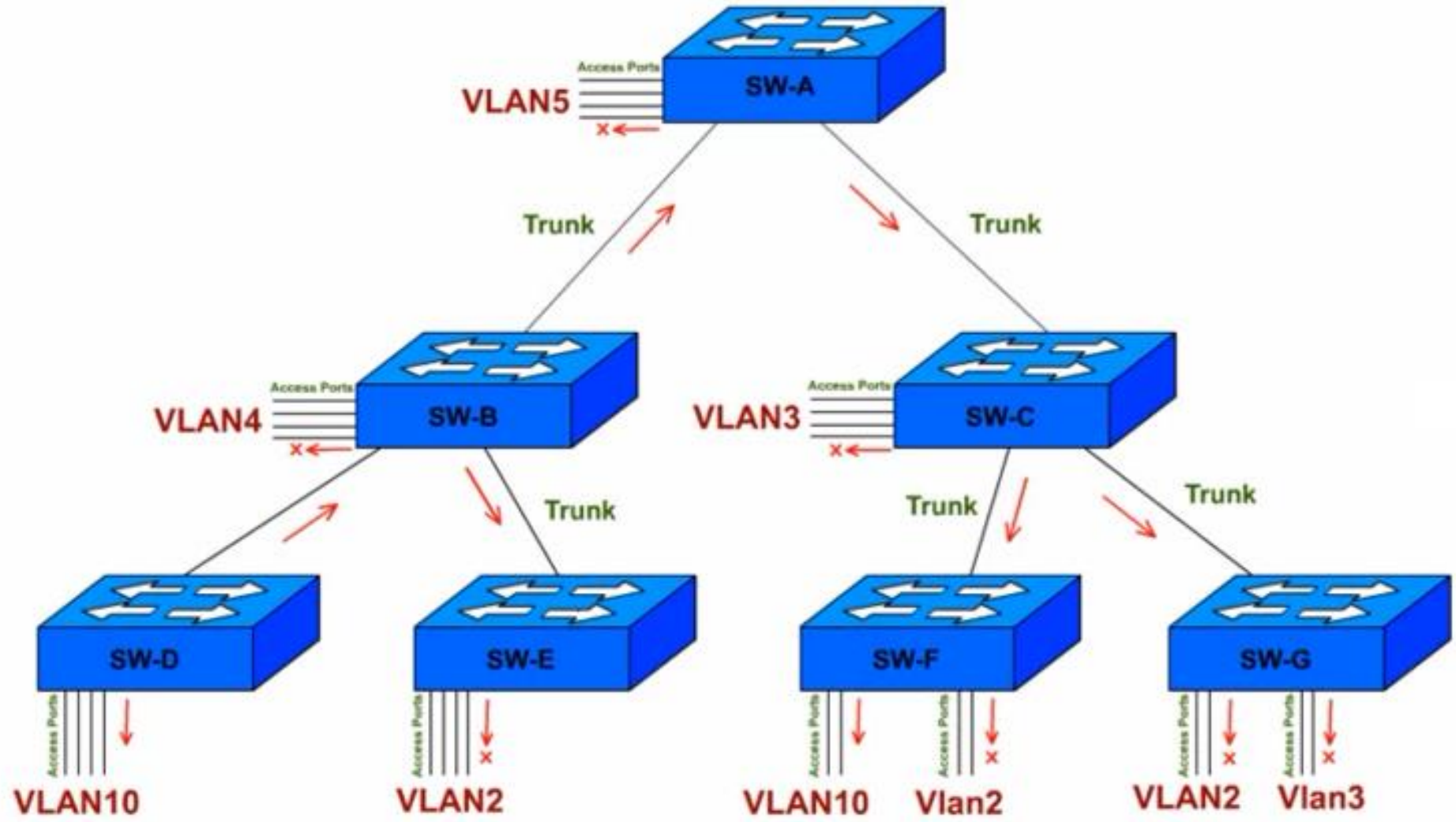
Configuration Commands	Where Stored	How to View
vtp domain	vlan.dat	show vtp status
vtp mode	vlan.dat	show vtp status
vtp password	vlan.dat	show vtp password
vtp pruning	vlan.dat	show vtp status
vlan *vlan-id*	vlan.dat	show vlan [brief]
name *vlan-name*	vlan.dat	show vlan [brief]
switchport access vlan *vlan-id*	running-config	show running-config, show interfaces switchport
switchport voice vlan *vlan-id*	running-config	show running-config, show interfaces switchport

VTP Configuration

Switch(config)# vtp mode [Server|Client|Transparent]
Switch(config)# vtp domain <*name*> (mandatory)
Switch(config)# vtp password <*password*> (optional)
Switch(config)# vtp version <*version*> (optional ,default version is 2)

Switch# Show vtp statues
Switch# Show vtp counters

VTP Pruning


VLAN5
Access Ports
SW-A
Trunk
Trunk
VLAN4
Access Ports
SW-B
VLAN3
Access Ports
SW-C
Trunk
Trunk
Trunk
SW-D
SW-E
SW-F
SW-G
Access Ports
VLAN10
Access Ports
VLAN2
Access Ports
VLAN10
Access Ports
Vlan2
Access Ports
VLAN2
Access Ports
Vlan3

VTP Pruning Configuration

Switch(config)# vtp prunning

Only on VTP Server